LITTLETON PUBLIC SCHOOLS
LIBRARY OF THINGS · ITEM GUIDE

1 setup

EinScan SE V2 3D Scanner

WHAT'S IN THE KIT

- EinScan SE V2 scanner with rotating turntable
- ThinkPad laptop with EinScan software
- Laptop charger

QUICK-START GUIDE

1. Place the object on the turntable, launch EinScan software on the included laptop.
2. Calibrate, then run a scan — the table rotates automatically.
3. Multiple scans merge automatically into one model.
4. Export as STL for 3D printing or editing in Tinkercad.

FULL LESSONS IN THIS GUIDE

PRO TIP: *Check out with the Bambu 3D printer for the complete scan-edit-print workflow.*

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LESSON · GRADES 6-8

Artifact Digitization Lab

LEARNING OUTCOMES

- Students will operate a structured-light 3D scanner
- Students will curate and document artifacts digitally
- Students will write interpretive museum text for an audience

MATERIALS

- EinScan SE V2 + turntable + included laptop
- Artifacts to scan: fossils, tools, sculptures, natural objects
- Exhibit card template (name, origin, significance)
- Shared drive or site for the digital museum

INSTRUCTIONS

1. Demo one full scan: placement, calibration, the rotating capture, the merged model.
2. Scanning crews rotate: each crew scans one artifact per session.
3. While waiting, crews research and draft their artifact's exhibit card.
4. Review models together: where did scans struggle (shiny, dark, thin parts) and why?
5. Assemble the digital museum: models paired with exhibit cards.
6. Launch: another class tours the digital museum and leaves visitor questions.

ACTIVITY

Artifact Digitization Lab — scan real objects into a curated digital museum with interpretive cards. The scanner's struggles with shiny and thin objects become the tech lesson.

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LESSON · GRADES 9-12

Scan, Remix, Print

LEARNING OUTCOMES

- Students will execute a full scan-edit-fabricate digital pipeline
- Students will modify a 3D mesh purposefully
- Students will evaluate fidelity loss at each pipeline stage

MATERIALS

- EinScan SE V2 + laptop
- Bambu 3D printer (checkout together)
- Tinkercad or Blender for mesh editing
- Objects with a story: a broken part, a favorite trinket, a tool to improve

INSTRUCTIONS

1. Students select an object and define the remix goal: repair it, extend it, or transform it.
2. Scan and export the STL; note surface quality issues immediately.
3. Import to Tinkercad/Blender; execute the modification.
4. Slice and print the remix on the Bambu.
5. Compare all four states: object, scan, edited model, print — where did fidelity drop?
6. Present the remix with a pipeline diagram and honest error analysis.

ACTIVITY

Scan, Remix, Print — the complete digital-fabrication loop on an object they chose. Tracking fidelity loss across each stage is the difference between using the pipeline and understanding it.